

MLA0409 - Deep learning① Machine Learning Vs Deep learning :-

- Machine learning (ML) is a branch of AI where algorithms learn Patterns from data to make Predictions Or decisions.
- Deep learning (DL) is a subset of ML that uses neural networks with multiple hidden layers.

<u>Aspect</u>	<u>Machine learning</u>	<u>Deep learning</u>
Architecture	uses algorithms like Decision tree, SVM	uses multi-layer neural networks such as CNN.
Data requirement	Can work with relatively smaller datasets	usually require large amounts of data.
Feature extraction	Features are generally selected manually	Features are automatically learned from data.
Training time.	Generally faster	Generally slower due to many Parameters
Applications	• spam detection • Credit risk Prediction	• Face recognition • AV's

* ML is suitable for many structured -data Problems.

* DL is complex data such as images, audio & text.

② Representation learning, width and Depth:-

* Representation learning is the process by which a neural network automatically learns useful features or representations from raw data instead of depending completely on manually designed features.

Ex:- Image recognition

Image $\rightarrow$ Edges $\rightarrow$ Shapes $\rightarrow$ Object Parts $\rightarrow$ Complete Object.

$\rightarrow$ The layers learn simple features, while deeper layers learn more complex features.

* Width refers to the number of neurons in a layer.

- more neurons allow the network to learn more patterns simultaneously.

- Excessive width can cause overfitting if the dataset is small.

* Depth refers to the number of layers in the network.

- more layers allow the network to learn hierarchical & complex representations.

- Increasing depth increases computational complexity & training time.

③. Relu, Leaky ReLU and ELU :-

* Activation functions introduce non-linearity into neural networks, allowing them to learn complex patterns.

ReLU :-

$$f(x) = \max(0, x)$$

- If $x > 0$, Output is x .
- If $x \leq 0$, Output is 0.

Advantages :-

- Simple & Computationally Fast.
- widely used in CNN's.

Use :- General deep learning & Computer vision.

Leaky ReLU :-

$$f(x) = \begin{cases} x, & x > 0 \\ \alpha x, & x \leq 0 \end{cases}$$

Where α is a small +ve value.

- +ve inputs behave like ReLU

Advantages :-

- Reduces the dying ReLU Problem.
- maintains a small gradient for negative inputs.

ELU :-

$$f(x) = \begin{cases} x, & x > 0 \\ \alpha(e^{x-1}), & x \leq 0 \end{cases}$$

- Positive inputs are linear
- Negative inputs smoothly approach $-\alpha$.

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Unsupervised Learning, RBM & Autoencoder :-

* Unsupervised learning is a learning method in which a neural network learns patterns and useful representations from unlabelled data.

Examples :- clustering, dimensionality reduction & feature learning.

* Restricted Boltzmann machine :-

An RBM is a Probabilistic generative neural network

Consisting of:

- visible layer :- Represents input data.
- Hidden layer :- Learning hidden features.

=> Input $\rightarrow$ visible layer $\rightarrow$ Hidden layer $\rightarrow$ Reconstruction.

* Autoencoder :-

An Autoencoder is a neural network that learns to reconstruct its input.

Working :-

Input $\rightarrow$ Encoder $\rightarrow$ latent Representation $\rightarrow$ Decoder $\rightarrow$ Output.

$$\text{Loss} = \|x - \hat{x}\|^2$$